

# MOHAMED ABO EL-SEOUD

Full-Stack Engineer Cairo, Egypt

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## SUMMARY

Backend Engineer with 2+ years of experience building scalable systems across **proptech**, **social media**, and **e-commerce** domains. Specialized in designing microservices, optimizing performance through caching and asynchronous pipelines, and managing cloud infrastructure on a daily basis. Focused on system reliability, clean architecture, and shipping production-grade solutions.

## SKILLS

**Tech Stack:** TypeScript, Go, SQL, GraphQL, NestJS, Express, React, Next.js, PostgreSQL, MongoDB, Redis Search, BullMQ.

**Tools:** Docker, Kubernetes, n8n, Datadog, NewRelic, OpenSearch, Sentry, Mixpanel, MoEngage, LangChain.

## EXPERIENCE

- Apr 2024 – Present **Full-Stack Engineer at Nawy** Cairo
- Engineered a **full-lifecycle CRM platform** supporting complex **multi-organization**, **multi-funnel** lead management with configurable **auto-assignment engines**, **lead delegation hierarchies**, and **agent performance rating** — enabling sales teams to operate across workflows with full traceability.
  - Reduced **database CPU utilization by ~32%** (from ~22% to ~15%) through two complementary strategies: introduced **Redis Search** for hierarchy-intensive queries — backed by **BullMQ** for async synchronization and **AWS Lambda** for continuous accuracy monitoring — and applied repository-wide **Redis caching** with dynamic cache invalidation across all lookup paths.
  - Architected a **multi-provider analytics service** using the **Strategy pattern**, abstracting event tracking across **Mixpanel**, **Google Analytics**, and **MoEngage** behind a unified interface for user identification, organization-aware context resolution, and comprehensive **user action tracking**.
  - Leveraged **AI** capabilities across the system, including automated **task quality evaluation** via **LLM** and **LangChain**, **lookup localization** through predefined **LLM prompt templates**, and **multilingual audio transcription** powered by **OpenAI Whisper**.
  - Designed **n8n workflows** for the auto-assignment service revamp, replacing hardcoded logic with **visual rule configuration** and conditional branching accessible to non-technical stakeholders.
  - Provisioned and configured **AWS infrastructure** via **Terraform** (ECS, API Gateway, Cognito, RDS) with hands-on implementation of advanced cases including **pre-Cognito token generation Lambdas**, **custom authorization flows**, and **verification Lambdas** for third-party service integrations, adhering to AWS security best practices.
- NestJS / PostgreSQL / Redis / AWS / Terraform / n8n / NewRelic / Datadog
- Aug 2023 – Apr 2024 **Backend Engineer at Baianat** New Damietta
- Delivered backend systems across **e-commerce**, **catering**, **e-learning**, and **social media** domains using **NestJS**, **GraphQL**, **PostgreSQL**, **MongoDB**, and **AWS**, with consistent patterns for auth, rate limiting, and error handling.
  - Built core real-time features including **live chat** with **Firebase push notifications** and **GraphQL Subscriptions**, and **shipping workflows** with geographic zone support.
  - Implemented **secure content delivery** via **S3 presigned URLs** and **Bunny CDN**, optimized **full-text search**, and async job processing with **BullMQ/Redis** queues.
- NestJS / GraphQL / PostgreSQL / MongoDB / Redis / AWS
- Mar 2023 – Jul 2023 **Backend Intern at Tuwaiq** El Sheikh Zayed
- Built and maintained **RESTful APIs** with comprehensive documentation, applying consistent patterns for **authentication**, **validation**, and **error handling**.
- NestJS / Express / PostgreSQL / REST API

## PROJECTS

- 🔴 **Sakan** Freelance – Sep 2025 – Feb 2026
- A marriage matching platform with full-stack ownership across frontend, backend, and deployment.
- Built **Micro-Frontend Architecture** using **Next.js** and **Chakra UI**.
  - Engineered **Automated Matching Pipelines** with **AI-Powered Chat Monitoring**.
  - Incorporated **Moyasar** for subscription-based payments and **Twilio** for messaging.
  - Deployed on **Alibaba Cloud** with dedicated **Worker Services** for matching, mail, and **SSE**-based real-time updates.
- Next.js / NestJS / OpenAI API / Alibaba Cloud / Twilio
- 🔴 **FRA Sandbox** Baianat – Nov 2024 – Feb 2025
- A fintech sandbox for **Egypt's Financial Supervisory Authority** to drive innovation.
- Built **Multi-Role Authentication** supporting innovators and enablers (accelerators, NBFIs, VCs).
  - Developed **Hackathon Management** with team discovery, role-based members, and invitation flows.

- Established **Real-Time Discussion Boards** with comments, replies, and notifications.
- Created **Live Ticketing System** for real-time support and issue tracking.

NestJS / PostgreSQL / TypeORM / GraphQL Subscriptions

## RIGOW

Baianat – Sep 2024 – Nov 2024

A social media platform for athletes and sports enthusiasts.

- Built **Real-Time Chat** with message status tracking (sent, delivered, edited).
- Crafted **AI-Driven Timeline** using **TensorFlow** for content recommendations.
- Established **Ranking System** prioritizing content based on user engagement.
- Applied typo-tolerant **Full-Text Search** for exploring topics and profiles.
- Created **Notification System** with localized alerts and optimized batch delivery.
- Enabled **Social Auth** via OAuth2 OpenID for Facebook, Google, and Apple.
- Used **Redis Pub/Sub** for real-time **GraphQL Subscriptions**.

NestJS / Redis / GraphQL / TensorFlow / PostgreSQL

## FASILA

Freelance – Jan 2024 – Mar 2024

An e-learning platform providing university students with affordable access to lectures and resources.

- Utilized **Worker Threads** to optimize CPU-intensive tasks.
- Ensured **Data Integrity** using **MongoDB Transactions** for purchases.
- Connected **Paymob** for seamless payments (card & wallet).
- Enforced **Single-Session Restrictions** to prevent account sharing.
- Secured content using **Presigned URLs**, CORS, and encryption.
- Configured **WhatsApp API** for cost-effective OTP delivery.
- Deployed **Dockerized services** on **DigitalOcean** with **Nginx Load Balancing** and **PM2**.

Express / MongoDB / Redis / Docker / DigitalOcean

## WORKFLOW

- **Performance Analysis:** Used **Datadog** and **NewRelic** to identify intensive endpoints, analyze slow database queries, detect lock contention, and drive targeted optimizations.
- **AI-Assisted Workflow:** Leveraged **Claude** and **Cursor** for accelerated engineering, including generating project rules, custom skills, and **MCP** tool integrations such as **Figma** and **Playwright** to streamline code generation, review cycles, and end-to-end testing.
- **Cloud & Infrastructure:** Worked daily with **AWS** services including **Cognito**, **SNS**, **SQS**, **ECS**, **API Gateway**, **Lambda**, **Secrets Manager**, and **RDS**. Used **Terraform** to provision Lambda functions and API Gateway configurations. Managed containerized deployments with **Docker**.

## EDUCATION

### B.Sc in Computer Science

Mansoura University

Offered a **Teaching Assistant** position — declined to pursue industry.

GPA: 91.1%

Sep 2020 – Jul 2024

## LANGUAGES

English - Professional, Arabic - Native

## CERTIFICATIONS

- **NTI:** Web Development (Node & React)
- **ITI:** Backend Development (Laravel & PHP)
- **CNBC:** 90% Most Common AWS Services
- **Udemy:** Troubleshooting Backend Performance